RoboCupJunior Rescue (Line/Maze) 2026
Team Description Paper

Tachyons

Abstract

Q-Botto (abbreviated as QB8) is a robot developed by the Tachyons team. Its main objective since the beginning of the project was to develop and introduce new technologies to the rescue maze competition, both for their educational value to the team's members and to differentiate the robot from other teams in the competition.

Its main chassis is CNC milled out of aluminum; it was designed by us and manufactured on a CNC machine lent by INFN LNS (the Italian National Institute of Nuclear Physics National Laboratories of the South in Catania), and features a custom PCB, designed by us and manufactured by JLCPCB. The components were soldered by hand.

Its motors are Stepper motors driven by TMC2209 stepper drivers for high reliability and good performance.

The robot's wheels are custom-designed and Injection molded with silicone by us.

The suspensions are a type of Christie suspension milled in aluminum in the same way as the chassis, with some 3d printed parts for parts with more complex geometries.

1. Introduction

a. Team

Our team is composed of four members:

- **Ettore Fins**: Team lead, Hardware and mechanics engineer and developer, Electronics and PCB designer and engineer, Software engineer, technical documentation author, test engineer. In 2025, he served as a key member of the MezaMaze rescue maze team, where he reached first place in the regionals and third in the national and European competitions. This year, he decided to form his own team to focus on the development of even more advanced hardware. He has designed all the electronics of the robot. He is a main part of the software development, and he designed the CAD of the current version of the robot and manufactured all the custom parts.
- **Francesco Sciuto**: Software developer, mapping algorithm author. In 2025, he served as the lead software developer and graphic designer for an On Stage Advanced team (Madagabot), which was awarded Best Technical Poster at the National Qualifiers and reached the European RoboCupJunior Championship. Building on this success, for the 2026 season, he transitioned his expertise into the Rescue Maze category, focusing on the MCU-side mission layer, including autonomous navigation, maze-mapping logic, and low-level sensor integration.
- **Angelo Caruso**: Graphic designer, technical documentation author. In 2025, he was part of an On Stage entry team (FermiTuttiTnT), where he was the main software developer and a jack of all trades for the construction of his main robot. For the 2026 season, he focused on the MPU-side software infrastructure, taking charge of the high-level Python dashboard, vision diagnostics, and the implementation of the embedded machine learning models for target recognition.
- **Gabriel Grasso**: Test and hardware engineer, woodworker, and labyrinth maker. This is the first time that he participates in Robocup Jr. He worked on building a wooden maze for testing purposes, and he designed some previous versions of the robot in CAD

2. Project Planning

a. Overall Project Plan

The project plan was driven by Rescue Maze rules, student time and budget limits, tests, and lessons from previous competitions. After the 2026 rule change added cognitive targets and Greek-letter victims, the plan moved from a single FOMO model to an Arduino UNO Q with OpenCV ROI and cognitive target detection, and YOLO for greek lettered victim recognition. Below a table of our requirements and tests of the main systems

Designed system	Minimum acceptable features and characteristics	Evidence for fulfilling requirements
Mechanical structure	Rigid chassis, stable suspension, traction and passive heat dissipation under use.	CAD, CNC manufacturing, Calculix modal/static simulations and physical maze tests.
ToF sensing	Left, front and right walls update consistently from VL53L7CX 8x8 matrices.	Bench validation, I2C multiplexer checks and maze tests.
Floor sensing	White, black, blue and silver/checkpoint samples classify from calibrated TCS3472 readings.	Calibration table, CRC validation, tile samples and maze runs.
Motion	One-cell movement, centering, 90-degree turns and repeatable stepper pulses.	Step-count validation, bench tests and maze runs.
Vision	Side ROI produces crops for Greek letters; ring reader extracts cognitive target colors.	OpenCV ROI tests, crop-classifier tests, replay/debug data and maze runs.
Rescue kits	Left/right deployment is repeatable and low-bounce.	Servo and mechanism bench tests and maze runs.
MCU - MPU Communication	Reliable bidirectional MCU - MPU communication of all necessary information	Dashboard status tests, map telemetry, and maze runs.

Constraints-based Robot and Team Requirements

Competition Rules and Control Strategy: The robot must be autonomous in its operation. All navigation, motor operation, sensor reading, color detection, and mapping functions are controlled entirely by the MCU. A manual switch must be used to start operations.

Physical and Dimensional Constraints: The maximum allowable height for the robot is 25 cm, and it must pass through maze tiles with minimum dimensions of 30×30 cm. The constraints demand a small enough space that 90-degree turns can be done without touching the walls.

Thermal and Structural Constraints: The stepper motors produce a heat and exert loads too high for a regular PLA 3D printed chassis or suspensions.

Electrical Constraints: High wire density results in untidiness and continuous disruptions due to vibrations from the robots.

Team Requirements: It is necessary for the team to design a customized PCB for the specific chassis to solve the problem of loose wires.

Sensory & Vision Constraints: Localization, wall detection, floor tile detection (black, silver, and colors), and victim identification are all important for the team.

Time & Budgetary Constraints: As high school students, it is difficult for the team to allocate much time since their days are spent at school. Thus, there were delays in ordering custom boards and using CNC machines. Financially, the team could not afford many changes. For example, we currently use cheap stepper motors because we have searched for sponsors to supply us with much more performant BLDC motors with gearboxes, but none could give us any in time, and given that we didn't have thousands of euros to buy them ourselves, we were left with the steppers

Rule function	Tachyons implementation	Status
Fully autonomous operation	The MCU owns navigation, mapping, movement and rescue-kit actions; the MPU only supports vision, logging, dashboard and evidence export.	Implemented and tested
Physical start control	The mission starts and stops from the A0 physical switch.	Implemented and tested
Wall detection and maze exploration	Three VL53L7CX 8x8 ToF sensors update left, front and right wall or ramp states in the internal maze map.	Implemented and tested
Black tiles	Floor sensors classify black tiles and trigger avoidance/recovery behavior.	Implemented and tested
Blue tiles	Blue floor classification is included in the floor-type pipeline and stored in the map state.	Implemented and tested
Checkpoints and Lack of Progress	The map stores pose, visited cells and mission state so checkpoint/LoP recovery can preserve mission knowledge.	Work is still being done to achieve sufficient reliability in distinguishing checkpoints from white or black tiles
Greek-letter victims	Side-camera ROI detection and ONNX crop classification identify Phi, Psi and Omega victims.	Implemented and tested
Cognitive targets	OpenCV ring analysis estimates the five ring colors and computes the victim status.	Implemented and tested, but additional work will be done to increase reliability.
Rescue kits	The MCU validates victim events and actuates the servo-based kit deployment mechanism.	Implemented and tested
Ramps, stairs and speed bumps	ToF and IMU evidence are stored in the 3D-capable map for traversal and replay.	Implemented and tested
Return / exit behavior	The navigation map supports exploration and return-to-start logic for end-of-run behavior.	Implemented and tested

b. Integration Plan

1. System Integration and Goal Attainment

For developing an efficient and entirely autonomous Robot, the team prioritized physical and electrical sturdiness prior to software implementation. Functions such as maze mapping and vision were not implemented until the physical hardware proved its validity.

The integration process hinged upon three pillars: tuning the 8×8 ToF matrices for accurate spatial mapping, guaranteeing mechatronics through switching from PLA materials to industry-grade hardware, and flexibility of algorithms for seamless adaptation to different competition floors.

2. Component Constraints and Fulfillment

All components were developed in accordance with constraints that stemmed from official rules of competitions:

Main Control Unit: Provides autonomous navigation, safety, motor control, sensor polling, map updates and rescue actions without requiring the dashboard.

Robot Chassis: Designed in Autodesk Fusion 360 according to restrictions of height up to 25 cm and maximum dimensions of 30x30 cm of a single tile, which allows 90-degree turns.

Aluminum Frame: The chassis, made of CNC-machined aluminum, was implemented to provide a structural passive heat sink for the stepper motors.

Sensors: The ToF matrices were responsible for wall mapping with noise filtering libraries, while color sensors recognized floor tiles based on CRC validation of calibration.

Custom PCB: Designed to achieve higher reliability and component density.

3. Internal Communication Architecture

The robot uses an MCU-centered architecture: the MCU owns mission decisions, while the MPU and Python application provide vision assistance, logging, playback and operator visibility.

The internal communication architecture is bidirectional but authority is intentionally asymmetric: the MCU reports state and accepts limited assistance messages, while the MPU cannot take over navigation.

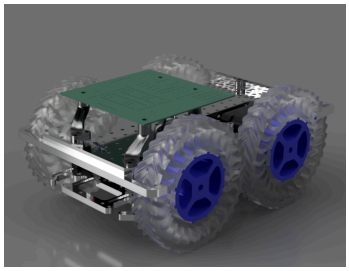
Input: The MCU takes inputs from the ToF matrices through proprietary libraries and color sensors through CRC checks.

Outputs and Diagnostics: The MCU communicates back with its internal map states and directs stepper motors to move ahead with tiles, and in parallel, secondary MPU modules run vision diagnostics for ROI victims.

3. Hardware

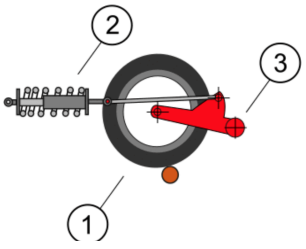
The hardware of the robot consists of the mechanical structure and the electronics. The structure is centered around a CNC-milled chassis. Above this main part, the PCB is held, with all the electronics except for the colour sensors, which are held beneath the chassis in front of the front motors. Above the Main chassis, the rescue kit deployment system is also held. It is an innovative design to shoot the cubes and make them hit the walls of the labyrinth without bouncing.

a. Mechanical Design and Manufacturing



Fusion 360 render of the main parts of the robot with a simplified PCB on top.

The main structural part of the robot is an aluminum chassis machined on a 3 axis CNC. It features an isogrid on the bottom to provide a high rigidity/weight ratio. In the intersection of different triangles, there are holes, as in a regular isogrid, but here the holes have an M3 thread to allow quick and robust mounting on the structure. The front motors are held between the chassis and a bottom plate, which are connected with some threaded rods to keep the motors under compression and in their correct position. The bottom plate also features two colour sensors for tile colour recognition. The back motors, instead of being held stationary like the front, are mounted to some Christie suspensions.



On the left, an image showing the wheel (1), the suspension lever (3), and the spring (2) is shown.

In our robot, the suspension lever is machined in aluminum in the same way as the chassis, and has on its rotational axis a threaded hole in which a threaded rod is tightly

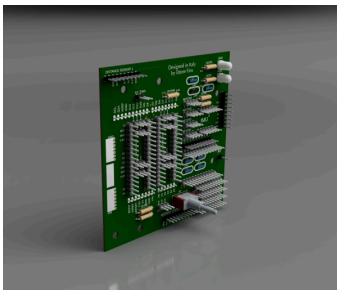
held, and which passes in some ball bearings held beneath the chassis to allow rotation. The spring is a tension spring, and it applies a force to make the lever go down and keep the chassis at its desired height. The motor is connected directly to the wheel and so moves up and down as well. The left and right suspensions are independent to allow navigation on difficult terrain.

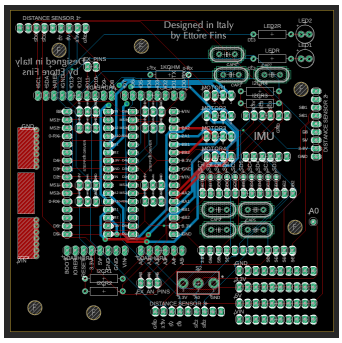
The wheels have been custom developed and injection molded by us, using regular transparent silicone caulk as the material to provide low cost and high traction. As can be seen on the Fusion 360 render of the robot above, the wheels are designed with an undulated surface, for higher traction on obstacles, and an airless design, for lower weight and rotational inertia, and higher dampening to avoid passing great transient shocks to the motors. The hollow parts of the wheel have some 3d printed TPU inserts, which have a much lower coefficient of restitution compared to the silicone, allowing for wheels with low losses during nominal operation (e.g. rolling on even terrain), but with a lowered coefficient of restitution on high loads, where the TPU parts experience high deformation. The wheels are molded with a translucent PETG center, which can easily be put on the wheel supports (blue parts at the center of the wheels seen in the Fusion 360 render above), for quick replacement in case of damage. The wheel supports are 3D printed in blue and red PLA and have a screw to allow a firm connection to the motor's shaft.

The rescue kit deployment system is unlike any other that we have seen in competitions. It consists of a set of rails with a slider held in the middle by two springs in tension pulling it on both sides. Pulling the slider to any side and releasing it makes it accelerate very rapidly towards the center. A servo motor is mounted on the center of the side of the rails. It rotates a compliant lever, which, when it reaches the center of the slider, has a pin that snaps into a channel in the slider. Rotating the lever, it moves the slider until the pin uncatches, and the slider is released and goes back to the center. The slider can house a Rescue kit on both sides of a central “bump”. Above the bump when the slider is centered, there is a tower that holds up to 8 kits, one above the other. When the bump is moved, a kit falls to one side of it, where it is then accelerated rapidly when the slider is released, allowing the cube to fly for about one meter. By rotating the lever to the right, a kit is shot left, and by rotating the lever to the left, a kit is shot right.

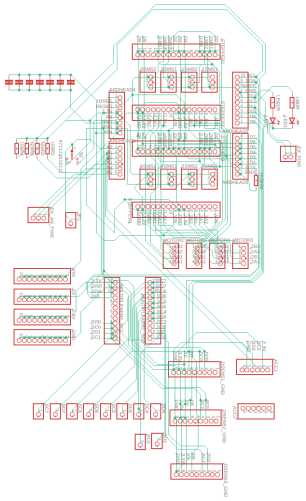
Our rescue kits are not mere 3d printed PLA cubes, but instead engineered with minimizing the coefficient of restitution as the main objective. They are mainly hollow with a wall thickness of about 0,8mm and a side length of about 10mm. They are 3D printed with a filament selected for the lowest coefficient of restitution, in this case, a blue TPU with unusual mechanical properties. On the inside, a mechanism is housed that dissipates kinetic energy during impacts.

b. Electronic Design and Manufacturing

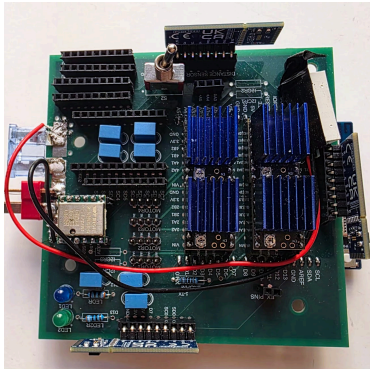




Our electronics design, like the hardware design, has the objective of high reliability. To this end, we have designed and manufactured (with JLCPCB) a custom PCB, Fusion 360 render below (note in the design of the pcb the components used are not exactly the same as the ones soldered to the real PCB because the correct part could not be found on publicly available libraries easily, for example in the design, quartz oscillators are placed instead of capacitors because the were the only component we had the design on with the correct pin spacing).



The PCB is a four-layered, so that the outer layers are used for signal traces and the inner layers are for large polygons, one connected to GND, and one connected to VIN in order to minimize interference and inductance. On the left is an image of the traces of layers 1 and 4, and on the right is an image of the electrical diagram. The PCB has all the electronic components of the robot, except for the color sensors, soldered, or, for the less reliable components, held by pinheaders for quick and easy replacement.



The main PCB components are:

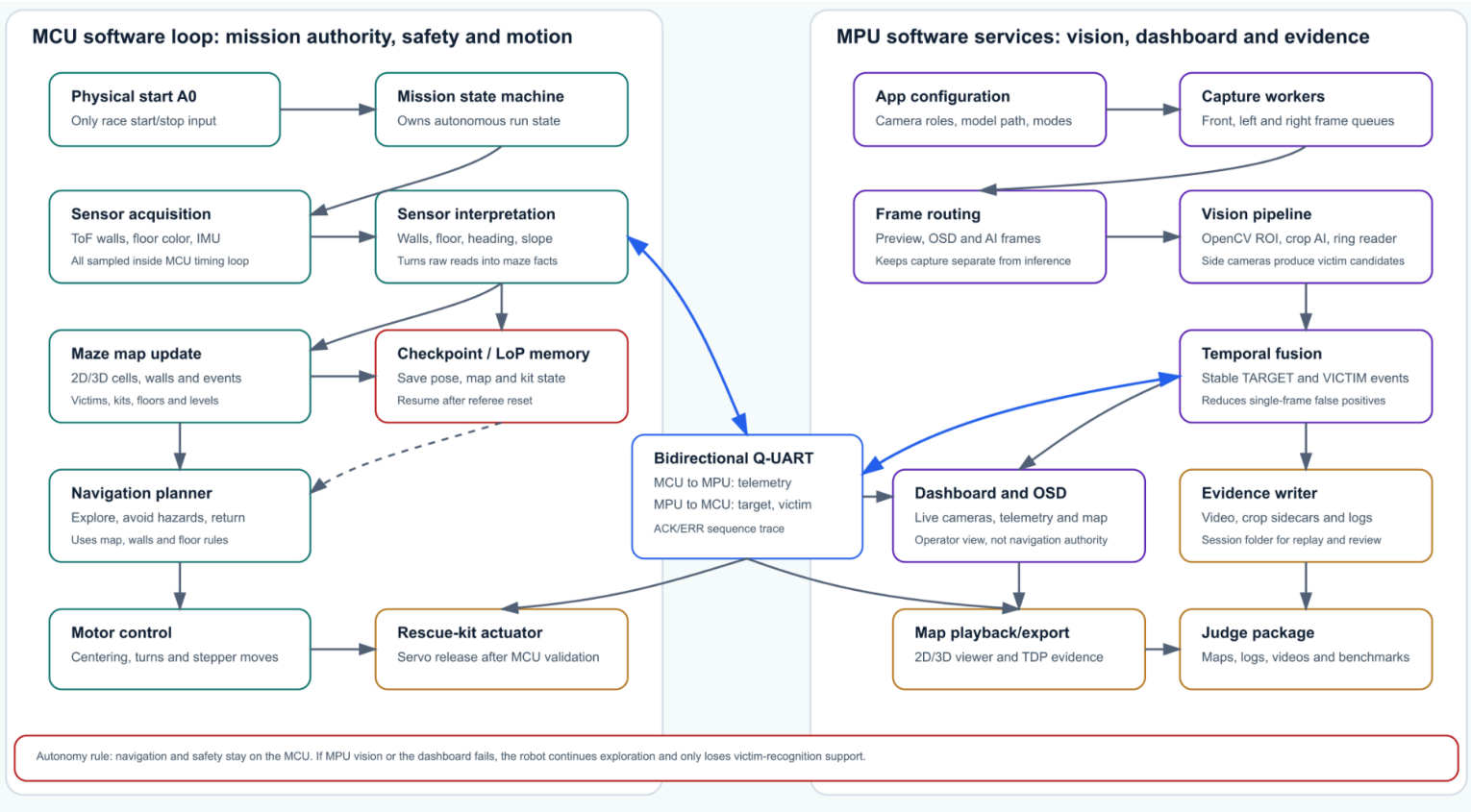
- VL53L7CX, an 8*8 matrix ToF distance sensor that we chose for its wide FoV, and which was given to us by STMicroelectronics.
- WT61P, an accelerometer and gyroscope that we chose for its precision and sensor fusion.
- TMC2209, a stepper motor driver that we chose for its low-vibration, operation, and UART communication, which allows us to read the load the motors are subjected to and change the motor coil current and speed accordingly.
- B32529C0564J000, film capacitors 560nf film capacitors for filtering and reducing noise on VIN.
- SW-T2-2E-A-A2-M1, switch for turning the robot on and off. In the render, it is the red switch.
- PCA9548A, the multiplexer for connecting all the I2C channels to a single I2C connection of the Arduino.
- Generic LEDs for indicating the status of the robot and identifying victims.
- Though not on the PCB, we have two TCS3472 colour sensors.

4. Software

The software of Q-Botto is designed around a clear separation between the autonomous robot controller and operator-support tools. Mission-critical code runs on the Arduino UNO Q MCU: it reads sensors, controls motors, updates the maze map, decides movements, handles safety states, and executes Rescue Maze actions. The MPU-side Python application supports camera setup, vision diagnostics, calibration, logging, map visualization and evidence export. The architecture was chosen so the robot can continue its mission if the dashboard, cameras or support layer are unavailable. The figure below shows the bidirectional Q-UART link and the key rule: navigation and safety remain on the MCU.

a. General software architecture

The software is organized into four main layers:



1. **MCU mission layer:** autonomous navigation, state machine, map, safety, motor control, ToF sensing, color sensing, and competition actions.
2. **Sensor and actuator layer:** VL53L7CX ToF sensors, TCS3472 color sensors, TMC2209 stepper drivers, mission switch, and rescue-kit actuator.

3. **MPU support layer:** Python dashboard, calibration tools, camera setup, video/log playback, and map visualization.
4. **Vision support layer:** side-camera target diagnostics using OpenCV ROI detection, an ONNX crop classifier for Greek-letter victims, and OpenCV ring analysis for cognitive targets.

The MCU software is structured around robot concepts rather than generic infrastructure. The main concepts are: robot pose, heading, cell, walls, floor type, map, navigation decision, motor command, and mission state. The mission starts from a physical input from the switch connected to A0. When the input is HIGH, the robot starts or continues the autonomous mission. When the input is LOW, the mission is reset and the motors are disabled.

The map is a compact grid of cells. Each cell stores its coordinates, level, floor type, wall state, target mask, visit count, and other mission flags. When the robot enters a cell, it marks the cell as visited. When it reads the ToF sensors, it updates the left, front, and right wall states. The wall updates are symmetric, so if one cell records a wall or opening, the neighboring cell is updated consistently.

The navigation algorithm currently follows a frontier-exploration strategy. It first tries to move forward, then left, then right, if the neighboring cell is open and unvisited. If all immediate neighbors are already visited or blocked, it uses a Dijkstra-style search on the visited map to find the nearest cell that still borders an unexplored frontier. This gives the robot a basic exploration and backtracking behavior without storing a large map.

The motor layer uses stepper motors driven by TMC2209 drivers. The MCU generates timed step pulses with acceleration ramps and tracks travelled distance through step counts. The robot is designed to move one cell at a time, in software, though in practice it moves fluidly and continuously, and only turns after it has stopped at the center of the current cell.

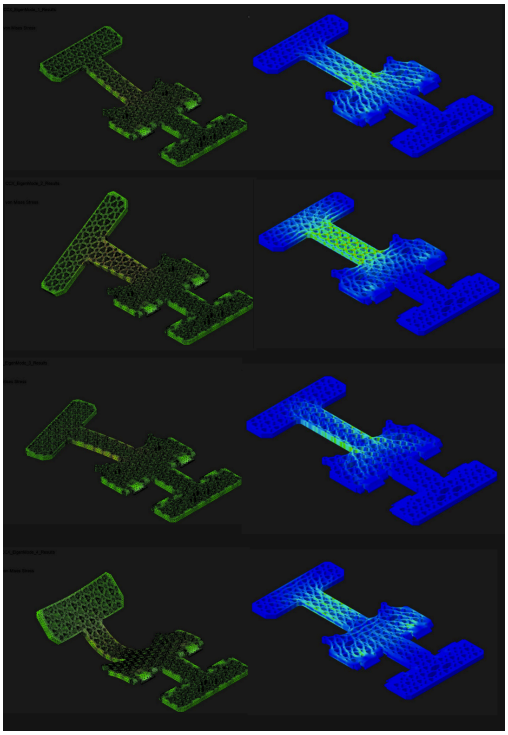
The sensor layer reads three VL53L7CX 8x8 ToF sensors through an I2C multiplexer. The raw matrix is reduced to distance values for the left, front, and right sides and wall inclination, used for PID motion, centering, and ramp recognition. Color is detected with two TCS3472 sensors mounted under the robot. The software compares normalized RGBC values against calibrated samples for white, black, blue, silver/checkpoint, and red tiles.

The Python support software provides a local dashboard. It shows mission state, sensor status, map state, camera views, calibration tools, and playback evidence. The dashboard can send color calibration data to the MCU through compact RouterBridge messages. It can also receive target events from the vision system, but the MCU remains responsible for the autonomous mission.

The left and right cameras are used for victim-target diagnostics. The vision software first searches for local regions of interest using OpenCV. For cognitive targets, it estimates the five colored rings and computes the health status from the ring values. For Greek-letter victims, the selected crop is passed to an ONNX crop-classification model. The model is not run on the full frame, which reduces CPU load and avoids treating the whole side-camera image as a victim candidate.

b. Innovative solutions

The vision system is also designed around a lightweight race strategy. The robot does not run full-frame AI as a normal behavior. Instead, OpenCV first finds possible target regions on the side cameras. Only those regions are analyzed for ring colors or passed to the YOLO model. This reduces computation, increasing FPS. Our robot also features many custom developed libraries for increased readability of the code and higher flexibility and customization compared to standard libraries.



5. Performance evaluation

Our robot has been put through a long and thorough set of tests. Starting from the design, the mechanical and electrical aspects have been simulated to ensure reliability. For example, we have done Calculix simulations to find stress concentrations and to ensure that our stepper motors do not operate close to the natural frequencies of our main chassis, as their vibrations could cause the structure to resonate (given the relatively high coefficient of restitution of aluminum)

On the left below, there are images of the four eigenmodes with 122,58 hz, 313,52 hz, 734,10 hz and 964,89 hz with their respective Von Mises stress on the right (the displacements are greatly scaled for easy visualization). The most important simulations were the static stress simulations that we used to find stress concentrations and displacement. We used the stress values to adjust the radius of the fillets, and the displacement value to choose the thickness of the aluminum chassis.

Once the robot parts were manufactured and we proceeded with integration, we did the actual test campaigns.
The majority of these were centered around our self designed and built labyrinth, see image below

Performance Evaluation Table. The integration tests are organized as measurable gates. For each gate, we record the test condition, result, and remaining risk.

Test gate	What we verify	Current evidence
Mechanical structure	Aluminum chassis rigidity, suspension behavior, wheel traction and passive motor heat dissipation	CAD design, CNC manufacturing, Calculix modal/static simulations and physical maze tests
ToF sensing	Left, front and right wall detection through VL53L7CX 8x8 sensors and I2C multiplexer	Bench validation and maze tests
Floor sensing	White, black, blue, red and silver/checkpoint classification with TCS3472 sensors	Calibration-based validation
Motion	One-cell movement, centering, 90-degree turns and repeatable stepper control	Step-count validation and maze tests
Vision	Greek-letter victim recognition on side-camera crops and cognitive-target ring analysis	OpenCV ROI tests and ONNX crop-classifier tests
Rescue kits	Left/right deployment, repeatable trajectory and low-bounce kit behavior	Servo and mechanism bench tests
Communication	Bidirectional MCU/MPU status, map, target and victim events	Q-UART dashboard tests
Mission reliability	A0 start/stop, autonomous continuation without dashboard authority, safe motor disable	Bench tests and maze tests
Evidence and replay	Logs, map export, playback and recorded diagnostic data	

2026 Rule Coverage Table

Rule function	Tachyons implementation
Fully autonomous operation	The MCU owns navigation, safety, mapping, movement and rescue-kit actions. The MPU only supports vision, logging, dashboard and evidence export.
Physical start control	The mission starts and stops from the A0 physical switch.
Wall detection and maze exploration	Three VL53L7CX 8x8 ToF sensors update left, front and right wall states in the internal maze map.
Black tiles	Floor sensors classify black tiles and trigger avoidance/recovery behavior.
Blue tiles	Blue floor classification is included in the floor-type pipeline and stored in the map state.
Checkpoints and Lack of Progress	The map stores pose, visited cells and mission state so checkpoint/LoP recovery can preserve mission knowledge.
Greek-letter victims	Side-camera ROI detection and ONNX crop classification identify Φ, Ψ and Ω victims.
Cognitive targets	OpenCV ring analysis estimates the five ring colors and computes the victim status.
Rescue kits	The MCU validates victim events and actuates the servo-based kit deployment mechanism.
Ramps, stairs and speed bumps	ToF and IMU evidence are stored in the 3D-capable map for traversal and replay.
Return / exit behavior	The navigation map supports exploration and return-to-start logic for end-of-run behavior.
Judge evidence	Dashboard logs, map playback/export and recorded evidence support post-run analysis.

6. Conclusion

The creation of Q-Botto by the Tachyons team was a meticulously integrated and designed hardware-software solution that was tailored to meet the specific physical and algorithmic challenges of the RoboCupJunior Rescue Maze competition. With the implementation of a 3-axis CNC milled aluminum chassis, which had been validated through Calculix modal and static stress simulation models, an optimal ratio of rigidity-to-weight was established, while the problem of thermal constraints presented by the hot stepper motors driven by the TMC2209 controller was solved by using the chassis as an efficient passive heatsink.

In terms of software and electronics, the electrical integration was made possible by our custom 4-layer PCB design that permitted a reliable, robust, and compact design. The MCU of the Arduino UNO was able to perform all the algorithms, sensor and motor controls, while communicating with the MPU for victim recognition. Through computation offloading through the use of efficient vision techniques involving the application of OpenCV region of interest (ROI) filtering before activation of the YOLO algorithm, the robot is able to maintain a high frame rate while identifying cognitive targets efficiently and effectively. Our thorough testing inside our proprietary physical testing maze system proves that we have addressed the past regional failure issues in terms of regional-stage execution. Overall, the creation of Q-Botto not only meets all the challenging autonomy criteria but is also a powerful learning pipeline in practicing engineering principles.